①

$$\frac{1}{1k} = 0.001A = 1mA$$

②

row 0 = 1V

Col 0 = 0V

$$\frac{V_{row0} - V_{col1}}{2k} = \frac{V_{col1} - V_{row1}}{1k} \quad (1)$$

$$\frac{V_{col1} - V_{row1}}{1k} = \frac{V_{row1} - V_{col0}}{2k}$$

$$= \frac{V_{col1} - V_{row1}}{1k} = \frac{V_{row1}}{2k}$$

$$= \frac{V_{col1} - V_{row1}}{1k} - \frac{V_{row1}}{2k} = 0$$

$$= \frac{2V_{col1}}{2k} - \frac{3V_{row1}}{2k} = 0 \quad \Big| \quad = \frac{V_{col1}}{1k} = \frac{3V_{row1}}{2k}$$

$= V_{col1} = \frac{3}{2} V_{row1} \quad (2)$

$$\frac{1 - \frac{3}{2} V_{row1}}{2k} = \frac{\frac{3}{2} V_{row1} - V_{row1}}{1k}$$

$$= \frac{1 - \frac{3}{2} V_{row1}}{2k} = \frac{\frac{1}{2} V_{row1}}{1k}$$

$$= \frac{1 - \frac{3}{2} V_{row1}}{2k} - \frac{V_{row1}}{2k}$$

$$= \frac{1}{2k} \frac{- \frac{5}{2} V_{row1}}{2k} = 0$$

$$= \frac{5}{2} V_{row1} = 1$$

$$V_{row1} = \frac{2}{5} = 0.4\ V$$

$$V_{col1} = \frac{3}{2}(0.4) = 0.6\ V$$

$$1mA + \frac{0.4}{2k} = 1.2mA$$

3

The floating nodes create unintended current paths that add error to the sensed output, corrupting the MVM result.
It gets worse in larger arrays because more floating nodes means more sneak paths and larger errors.